

महाराष्ट्र वन सेवा मुख्य परीक्षा - 2024 दिनांक - १३ मे, 2024



2024

D20

BOOKLET NO.

280026

**Forest Services
Statistics**

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Questions no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – 1. Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

SEAL

**SECTION – A****Q1.** Solve **any five** out of seven.**(8×5=40)**

- (a) A study of Airline delays shows that 40% of all the days have rainy weather, while the remaining 60% are clear. On a rainy day 30% flights are delayed. On a clear day 10% flights are delayed and overall 15% of all flights are delayed. Given that a flight was delayed. What is the probability that it was a rainy day ?
- (b) A researcher is studying the mean (μ) of a normal distribution. She processes using sample mean ($\bar{x}$) as an estimator for μ . Given that sample is drawn from a population with $x_1, x_2 \dots x_n \sim N(\mu, \sigma^2)$ then
- Prove that $\bar{x}$ is a unbiased estimator for μ .
 - Prove that $\bar{x}$ is a consistent estimator for μ as $n \rightarrow \infty$.
- (c) A teacher wants to predict a students exam score based on number of hours they studied. Below is data collected from a group of 5 students regarding the hours studied and exam scores.

Hours studied	Exam score
1	55
2	60
3	65
4	70
5	75

Use the Least Square method to fit a linear regression model to predict the exam score based on the number of hours studied.

- (d) A researcher is conducting a study on the annual income of the employees in a large company. The company has 10,000 employees with 3000 working in the marketing department, 4000 in finance and 3000 in HR department. The researcher plans to use 400 employees for survey.

How many employees should be selected from each department using simple random sampling based on the proportion of employees in each department ?



- (e) Consider a hypothesis testing problem for a normal distribution. Let $x_1, x_2, \dots, x_n$ be iid random variable with following properties.

i) $x_i \sim N(\mu, 1)$ where μ is the unknown mean

where

Null hypothesis : $H_0 : \mu = 0$

Composite hypothesis : $H_1 : \mu \neq 0$

Find the test statistics for this hypothesis testing problem.

- (f) State the Gauss-Markov theorem and its implication.
 (g) Discuss three different types of sampling techniques, that the researcher could use to select the sample and explain how each method work ?

- Q2.** (a) i) Let X be a non negative random variable with $E[x] = 10$. Using Markov's inequality, find an upper bound for $P(x \geq 20)$. **15**
 ii) Let X be a random variable with $E[X] = 5$ and $\text{var}(X) = 4$ using Chebyshev's inequality. Find an upper bound for $P(|X - 5| \geq 3)$.
 (b) State and prove Lehmann – Scheffe theorem. **15**
 (c) Let $X \sim W_p(\Sigma, n)$ be a Wishart – distributed random matrix. Show that $y = X^{-1}$, then $Y \sim \text{inv-wishart}(\Sigma^{-1}, n)$. **10**

- Q3.** (a) Consider a non linear model given by the equation $y = \beta_0 + \beta_1 x_1^2 + \beta_2 x_2^3 + \epsilon$. Where ϵ is the error term. **15**
 i) Perform a sensitivity analysis to understand how changes in the input variables x_1 and x_2 influence the output y .
 ii) Also can you quantify the effect of each input variables on the output using partial derivatives.
 (b) Consider an experiment designed to study the effects of two factors A and B each at two levels in a 2^2 factorial design. The experiment consists of four runs with the following result. **15**

Run	Factor A	Factor B	Responses (Y)
1	-1	-1	5
2	-1	1	7
3	1	-1	9
4	1	1	-11

Calculate the main effect of factors A and B. How would you interpret the interaction effect in this context ?

- (c) If X and Y are two random variables and suppose $E[X^2] < \infty$ and $E[Y^2] < \infty$ where $E[\cdot]$ denotes the expectation operator. Using the Cauchy-Schwarz inequality prove that $|E[X, Y]| \leq \sqrt{E[X^2]E[Y^2]}$. **10**



- Q4.** (a) A researcher is studying the relationship between gender and smoking behaviour. A survey was conducted on 200 individuals and result are as follows :

Gender/Smoking	Smokers	Non smokers	Total
Male	40	60	100
Female	30	70	100
Total	70	130	200

Use Chi-square test at 5% significance level to determine if there is a significant association between gender and smoking behaviour. (For $\alpha = 0.05$ and $df = 1$, χ^2 value = 3.841) **15**

- (b) Let there be a balanced incomplete Block design with parameters (v, k, λ) where v is the number of treatments (Elements), k is the number of treatment in each block and λ is the number of times each pair of treatments appears together in the blocks. Show that following relationship must hold :

i) $b = \frac{v(v-1)}{k(k-1)} \lambda$

where the number of blocks b in BZD is related to v , k and λ .

- ii) The total number of pairs of treatment $\left(\frac{v}{2}\right)$ is equal to the total number

of pairs in the blocks. Which can be expressed as $\lambda \cdot \left(\frac{k}{2}\right) \cdot b$? **15**

- (c) A service centre has three types of customer service states for their representatives. Trainee(T), Experienced(E) and Expert(x). The service centre has observed the transition between three states over time and these transition follows a Markov Chain. **10**

The Transition matrix is given by

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$$

Where the states are represented in the order : Trainee, Experienced, Expert.

The row represents current state and column represents next state.

Find the steady-state distribution of the customer service representatives.

**SECTION – B****Q5. Solve any five out of seven.****(8×5=40)**

- (a) Define
- Survival function
 - Hazard function
 - Cumulative Hazard function.
 - Cumulative distribution function.
- (b) Write short notes on dual problem when primal is in standard form.
- (c) Consider general linear model $Y = X\beta + e$ where, $e \sim N(0, \sigma^2 I)$, Y is the $n \times 1$ vector of n observations, X is $n \times p$ design matrix, β is $p \times 1$ vector of parameters, e is $n \times 1$ vector of random errors.
- Show that $X' \times \hat{\beta} = X' Y$
 - Show that $\hat{\sigma}^2 = \frac{Y'(I - P)Y}{n - P}$
where, P is not matrix.
- (d) Write short notes on following terms :
- Standard scores.
 - T – scores.
 - Validity and test length.
- (e) Write short note on type of inventory and cost involved in inventory problems.
- (f) Write short note on :
- Time reversal test.
 - Factor reversal test.
- (g) Define :
- General Fertility Rate (GFR)
 - Total Fertility Rate (TFR)
 - Standardised Death Rate (STDR)
 - Infant Death Rate (IDR)



Q6. (a) i) Explain in details nature of hazard function. **15**

ii) Suppose a life time r.v. has density function

$$f(t) = \beta t^{\beta-1} e^{-t^\beta}; t > 0, \beta > 0$$

$$= 0 \quad ; \text{ o.w.}$$

Obtain hazard function and also identify nature of hazard function.

(b) i) Solve the following assignment problem. **7**

	1	2	3	4
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

ii) Solve the following problem to maximize profit. Which job is to be left? **8**

		Job				
		A	B	C	D	E
Mechanic	1	62	78	50	101	82
	2	71	84	55	73	59
	3	87	92	111	71	81
	4	48	64	87	77	80

(c) Write short notes on **(Any two)** : **10**

i) Laspeyre's index number.

ii) Paasche's index number.

iii) Fishers index number.

Q7. (a) A popular consumer co-operative store located in a labour colony reported the average monthly data on prices and quantities sold of a group of selected items of mass consumption as given below : **15**

Compute :

i) Laspeyre's price index for 2012 using 2008 as the base

ii) Laspeyre's price index for 2008 using 2012

iii) Paasche's price index for 2012 using 2008

iv) Paasche's price index for 2008 using 2012

v) Fisher's quantity index for 2012 using 2008



vi) Fisher's quantity index for 2008 using 2012

Items	January 2008		January 2012	
	Prices (Rs. per kg)	Quantities sold (kg)	Prices (Rs. per kg)	Quantity sold (kg)
Veg oil	26	40	31	45
Pulses	28	90	32	100
Jaggary	16	120	19	20
Rice	15	110	18	130

(b) Complete the following life table.

15

Age in Yrs.	l_x	dx	qx	P_x	L_x	T_x	ex^0
11	14,000	—	—	—	—	—	—
12	10,000	—	—	—	—	—	—
13	8,000	—	—	—	—	—	—
14	6,000				4,000	—	—

Where, l_x – The number of persons surviving to the exact age x years.

dx – No. of persons dying between exact age x and $x + 1$.

qx – The rate of mortality at age x .

P_x – The probability that a person aged x will survive till the age of $(x+1)$ years.

L_x – The average number of person of a stationary population living between ages x and $x + 1$.

T_x – The total number of years lived by the cohort after attaining an exact age x .

ex^0 – The complete expectations of life at age x .

c) Write short notes on **(Any two)** :

10

i) Single sampling plan.

ii) Double sampling plan.

iii) Average Outgoing Quality curve (AOQ).



Q8. (a) A super market has two girls ringing up sales at counters. If the service time for each customer is exponential with mean 4 minutes and if people arrive in a Poisson fashion at the rate of 10 per hr. **15**

- i) What is the probability of having to wait for service ?
- ii) What is the expected percentage of idle time of each girl ?
- iii) If customer has to wait what is the expected length of his waiting time ?

(b) Compute the standardized death rate by Direct method and Indirect method of standardization for the following data : **15**

Age group	Population State A	Deaths in State A	Standard Population	Standard Death rate
0 – 5	1,000	50	10,000	70
5 – 10	900	22	9,000	40
10 – 25	1,100	14	11,000	20
25 – 45	3,000	15	30,000	06
45 and above	4,000	55	40,000	12

(c) Write short note on **(Any two)** : **10**

- i) Average Sample Numbers Curve (ASN)
 - ii) Average Total Inspection Curve (ATI)
 - iii) Operating Characteristics Curve (OC)
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